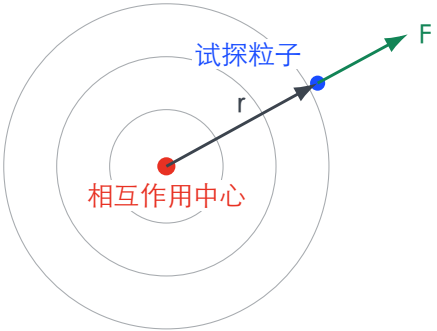
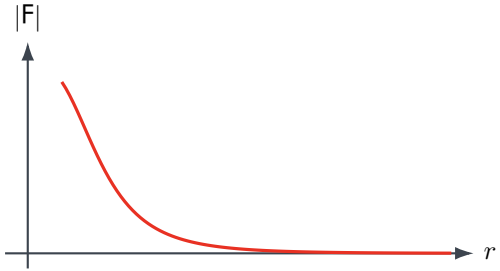
汤川型势的保守力：方向径向向外，近区类库仑，远区指数屏蔽



$U_0 > 0$ 且 $dU/dr < 0$，故 $-\nabla U$ 沿 $+\hat{\mathrm{r}}$，表现为排斥。



$$\mathsf{F} = U_0 e^{-\alpha r}\left(\frac{1}{r^2}+\frac{\alpha}{r}\right)\hat{\mathrm{r}}$$

$r \ll \alpha^{-1}$：$|\mathsf{F}| \sim U_0/r^2$；
$r \gg \alpha^{-1}$：$|\mathsf{F}| \sim \alpha U_0 e^{-\alpha r}/r$。